Priyesh Gupta

AI & Quant Systems Engineer | Founder, AlphaCalendar

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SUMMARY

AI and quant systems engineer who builds end-to-end data, machine-learning and LLM products. Founder and sole engineer of AlphaCalendar (alphacalendar.in), a live research platform for Indian equities: 171 versioned data contracts, an 84M-row point-in-time data lake, LLM extraction over 9,497 earnings calls and a cited RAG assistant at ~$0.0013 per answer. Former applied-AI researcher at IIT Bombay and IIT Madras with 4 peer-reviewed papers, including the IEEE TALE 2024 Best Student Paper.

SKILLS

Languages: Python, C++, Rust, Go, TypeScript, SQL, JavaScript, R

AI / LLM: LLMs, RAG, embeddings, vector search, tool calling, structured outputs, OpenAI API, Gemini, Qwen, LLaMA, llama.cpp, FAISS, LangChain, Hugging Face, Whisper, fine-tuning

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, forecasting, NLP, speech, computer vision, SHAP

Quant & Statistics: Walk-forward backtesting, point-in-time data, permutation tests, Benjamini-Hochberg FDR, options pricing (Black-76, Black-Scholes, SVI, Greeks), Monte Carlo, ARMA/GARCH, DTW

Data Engineering: Polars, pandas, NumPy, SciPy, DuckDB, Parquet, PostgreSQL, ETL pipelines, data contracts, data validation, XBRL parsing, web scraping

Backend, Web & Infra: Next.js, React, Node.js, FastAPI, asyncio, REST APIs, WebSockets, Linux, Cloudflare, Oracle Cloud, Supabase, Git, GitHub Actions

EXPERIENCE

Founder & Sole Engineer, AlphaCalendar (alphacalendar.in) May 2025 – Present

- Built and launched (Oct 2026) a research platform for Indian equities end to end: data pipelines, quant models, LLM features, Next.js app, payments and infra.
- Architected a contract-first, point-in-time data platform of 171 versioned datasets (~84M rows) with 15 validation gates, SHA-256 lineage and lookahead prevention enforced in code.
- Shipped LLM extraction over 9,497 earnings calls (1,470 companies), turning guidance into 44,610 tracked management promises; built a cited RAG assistant with int8 embeddings and read-only tools at ~$0.0013 per answer under a hard spend cap.
- Ran 20,851 seasonality tests with permutation testing and Benjamini-Hochberg FDR control (936 survived).
- Cut the 6M-row feature build from 820 s to 147 s with byte-identical output; run the site and pipeline on a single free ARM VM behind Cloudflare Tunnel at ₹0 hosting cost.

Co-founder & Chief Data Engineer, BazaarBuzz Jan 2025 – 2026

- Built on-device transcript intelligence with Qwen-2.5-7B, FAISS and sentence-transformers, processing a full Nifty-500 earnings season in 2.4 hours on one consumer GPU with no external APIs.
- Engineered an event-driven execution platform (asyncio, Pydantic) with a 17-point pre-trade risk gate: 18 ms average risk check and zero orphan orders across 42 simulated WebSocket disconnects.

Research Intern, IIT Madras Jun 2024 – 2025

- Researched multimodal stock prediction fusing transformer sentiment, news, technicals and fundamentals.

Research Associate, Educational Technology, IIT Bombay May 2023 – Aug 2024

- Improved speaker diarization for Indo-English classrooms with Whisper and PyAnnote across 140+ hours of interactions; won Best Student Paper at IEEE TALE 2024.
- Built multimodal emotion recognition (88% accuracy) and CRF models for regulation triggers (T4E 2024, T4E 2025).

Full-Stack & MLOps Intern, Vocab.AI Aug 2023 – May 2024

- Fine-tuned LLaMA-2 and GPT-3.5 Turbo for sentiment and conversation scoring; shipped diarization and voice-emotion models (81% accuracy) via REST APIs and React.

Research Intern, NIT Puducherry Dec 2023 – Jan 2024

- Built energy-efficient CNNs for lung-nodule segmentation (77% more efficient); published at IEEE INDISCON 2025.

EDUCATION

WorldQuant University, MSc in Financial Engineering (in progress) Sep 2025 – Present

Indian Institute of Information Technology (IIIT) Dharwad 2021 – 2025

B.Tech, Data Science and Artificial Intelligence; CPI 8.06/10

PUBLICATIONS & AWARDS

- Advancing Speaker Diarization With Whisper Speech Recognition, IEEE TALE 2024 (Best Student Paper)
- Energy-Efficient Depthwise Separable CNNs for Lung Cancer Nodules, IEEE INDISCON 2025
- Emotional Trajectories through Valence and Arousal in Math Learning, T4E 2025 (APSCE)
- Automating Identification of Socially Shared Metacognitive Regulation Triggers, T4E 2024 (Springer)
- IEEE INDISCON 2025 Reviewer; UG Fellow, TIH FinTech (IIT Bhilai); 1st, DRS Hackathon; Top-15, HackFest NITTE